Citation (APA)	Research Question	Methodology	Key findings	Gaps/limitations	Relevance to my project
(de Sá-Freitas et al., 2021)	To determine the correlation between star formation rates and galaxy morphology	Experimental: Data and used a quantitative analysis using star formation acceleration (SFA)	That Elliptical galaxies quenched faster than spiral galaxies	Pointed to some likely contenders for mechanics behind the quenching star formation in elliptical galaxies but was not specific.	Could find a potential link between supernovas in elliptical host galaxies and location. Provide more evidence to the mechanics of galaxy evolution.
(Möller et al., 2021)	To develop a broker that can provide a steady access to ZTF and LSST data that has been pre-processed and classified, streamlining analysis.	Experimental: Data is ingested from telescope streams and stored on machine disks, data is passed through science modules which do filtering and classification and matching.	Fink allows for easier follow up on transient objects such as supernova, microlensing events and gamma ray burst due to its rapid processing and classification of data from other telescopes.	Technical paper, beyond my scope of understanding the technical proceedings. A gap and or limitation for Fink is the science modules and filters only working for certain objects and the machine learning models performing poorly.	I will use this broker to access LSST data to find the correlation between the supernova position and the properties of the host galaxy. It is important that I understand some of the technical side and it help contextualise the main tool of my project.
(Deckers et al., 2022)	To fit light curves from 115 objects from the ZTF 2018 sample to a model grid that had various distributions of 56Ni shells with 67 % of them being explained as Chandrasekha	Fitting experimental data from ZTF 2018 sample to the TURTLS model grid and attempting to constrain parameters to match light curves in the g and r band.	The supernovas fit to the TURTLS model grid of Chandrasekhar mass explosion with a range of 56 Ni distributions, masses and explosion energies. With a preference	Lacking data specifically for the early flux excess SNe Ia to make conclusions regarding correlations about detection efficiencies. Encourages the use of rapid	I suspect Anais will ask me to focus on the early excess SNe Ia light curves as Rubin LSST will collect a lot of data with potentially more of these being found and

	r mass explosions. Also attempted to explain the early excess fluxes appearing in 6 SNe Ia by using the colours (g and r bands) and the lifetimes as explained by companion, CSM and 56Ni clump models.		for extended 56Ni distributions. 6 early flux excess SNe Ia are identified in the gr band data and they find with 18+-11 % of finding early flux excess SNe Ia at z < 0.07 and a correlation with lower galaxy mass but this was not statistically significant due to a lack of data.	detection of these SNe so spectroscopic follow up can occur which enables for progenitor mechanics to be thought about.	supported by follow up spectroscopic observation. This will help greatly in the identification of the specific type of supernova and eventually I will investigate host properties of the galaxies with these supernovae.
Ramaiya et al., 2025	Finding possible explanations to the colour – luminosity relation of SNe Ia with host galaxy properties.	Experimental: Used data for SNe Ia from DES and host galaxy data from a variety of telescopes and then plotting the SFR against the stellar mass with the reddening of the host galaxy as well as splitting it into three regions of high mass bursting, low mass bursting and high mass passive galaxies.	That the slope of the SN colour luminosity relation is dependent of the region of the SFR vs Stellar mass plot based on the galaxy properties. As well as that SNe Ia are 0.07 to 0.11 mag brighter in High mass passive galaxies.	Small data issue seem to plague these studies as well as the methods used to determine the host galaxies of the SNe, although I didn't read too deeply into methods. The paper specifically mentioned the relevance of the LSST using Rubin due to the large amounts of SNe that it's expected to discover.	Literally is a study that I can use numbers and values from to compare to my research to understand my own results and how it fits into the context of the current research and literature. I suspect this will be one I return to.
(Srivastav et al., 2023)	Find appropriate constraints	Experimental: Using photometric	(Srivastav et al., 2023) ruled out ejecta	Limited access to follow up data	As the scope of my project narrows I am

	using a MOSFiT model of hybrid 02es-like type Ia supernova to rule out the mechanisms of the explosion.	and spectroscopic data fit to a model for transient objects to classify the constraints for these curves to occur.	companion interaction, massive helium shell detonation and 56Ni distribution being drivers of the explosion and emphasised that the most reasonable explanation was the interaction of the SNe with the circumstellar medium.	limited the ability to place further constraints on the MOSFiT modelling to understand the cause of the circumstellar medium whether it was from tidal tails caused by WD merger event or disk originated matter caused by a disruption to the secondary white dwarf.	looking for specific supernovae that are interesting to study, 02es types seem to have this early excess bump that I am interested in and this paper will help me in understanding the properties that lead to these early excess's'.
(Leoni et al., 2022)	To demonstrate how an active learning loop can be an effective method to sample a new training dataset for a classification model given small validation data.	A pipeline of filtering, feature extraction and then classification to create the sample that the true model will be retrained on.	That an active training regime using uncertainty sampling can be an effective method to create training datasets that can be used to retrain a classification model like that on Fink. This active learning regime achieved a purity of 89 per cent and an efficiency of 54 per cent using uncertainty sampling.	Limitations are I think in the classification of other transients as the other classes were broad brushed such as AGN-like and other_SN as well as how this will be effectively implemented into the retraining of the Fink Supernova classification module.	This gives me more context to how Fink works when classifying the data that I will be using for my project. I need to be able to understand where the data comes from and how it is processed to analyse it effectively.
(Anderson et al., 2015)	Using different bands flux data for 102 star forming host galaxies to	Experimental: Fitting supernovas occurring in different	That SNe Ia are occurring in the blue band of these host galaxies which	No consideration of how dust affects results and limited	This shows how the type of SN that I am searching for is more

	trace out what band types of supernovae are appearing in. This leads to the constraints that can be placed on the time delay distribution between when the explosion happens and what the progenitor was.	bands of light from host galaxies using a pixel statistic NCR method. Host galaxies emit light from different parts of the galaxy so finding what band the supernovae are occurring in can find a spatial correlation with the host galaxy.	shows that the progenitor of the SNe Ia cannot have a tiem delay of less than 100 Myr. Because otherwise you would see the supernovae fitting to the cumulative distribution in the H alpha and UV bands. But they did not fit to these bands.	conclusions on constraints placed on progenitor channels.	likely to be found in the blue bands of host galaxies, this will be crucial in developing the sample of SNe that I will search for Early Excess SNe in as these Early Execess SNe only occur in type Ias
(Rigault et al., 2020)	To demonstrate how the local environment around a star and the age of the progenitor play a greater role in the standardisatio n of the brightness of the SNe.	Experimental: Using data from the SDSS and the nearby SN factory using 141 SNe Ia and then categorising the SNe by age and deriving the local specific star formation rate which is the new method used to probe the local environment of these supernovae.	The results show that fitting these SNe Ia on plots of colour, stretch and Hubble residuals as a function of the lsSFR shows a correlation with the stretch with a pearsons of -0.48 with a significance of 5 sigma, no correlation with colour and also finding that the mass step discrepancies appears when plotting Hubble residuals as a function of lsSFR. Accounting for biases in the	No ties to progenitors of the SNe Ia were mentioned and how the lsSFR could be used in that context, although this paper was determined to demonstrate the effectiveness of reducing the step in the Hubble residuals.	This paper cleared up the calibration process that occurs when analysing the light curve of SNe Ia and how calibration for Ia depends more on the intrinsic properties of the SN and less on the extrinsic properties. Hence the importance to understand the explosion mechanisms despite what the author of the paper suggests.

			lsSFR and global stellar mass biases reduces the mass step by 70 per cent of its original value. Showing that the lsSFR which points to that this mass step is related to an intrinsic property of the SNe Ia and not something unrelated like the stellar mass of the Host.		